Multi-Scale Frequency Pyramid Fusion with Hierarchical Attention for Explainable Medical Image Classification

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Introduction & Motivation

- Medical image classification is critical for automated diagnosis across modalities like **MRI** and **chest X-ray radiography**.
- Standard CNNs operate on spatial-domain inputs and may miss discriminative **high-frequency spectral patterns** (edges, textures, fine boundaries).
- The **Fast Fourier Transform (FFT)** provides a complementary frequency-domain perspective, decomposing images into magnitude and phase components.
- **Key Insight:** Integrating FFT-based representations at a *single* network depth (Layer 4 only) captures only one semantic level, missing multi-scale pathological characteristics.
- **Our Approach:** Multi-Scale Frequency Pyramid Fusion (MSPF) — applying frequency-aware processing across *multiple* network depths and fusing via a Feature Pyramid Network with learnable gating.

Literature Review

Author	Description	Methodology	Results	Limitations
Xiao et al. IEEE	**Multi-Scale Frequency Feature Fusion Network (MFN) for multi-class segmentation.**	**Refined ASPP with fine-grained atrous conv, hybrid attention, and FreqFusion for feature smoothing.**	**Demonstrated superior performance on Cityscapes and Pascal VOC datasets.**	**Standard multi-class methods suffer from boundary ambiguity and detail loss (addressed by MFN).**
Chen et al. IEEE	**Frequency-aware Feature Fusion (FreqFusion) for dense image prediction tasks.**	**Adaptive Low-Pass Filter (ALPF) for consistency, offset generator for resampling, and high-pass filter (AHPF) for boundaries.**	**Cityscapes: +2.35% mIoU. COCO Object Detection: +1.7 AP (box).**	**Simple initial fusion in standard methods causes blurring; the method addresses this, but adds complexity.**
Feng et al. IEEE	**Context Pyramid Fusion Network (CPFNet) for medical image segmentation (e.g., skin lesions).**	**Global Pyramid Guidance (GPG) and Scale-Aware Pyramid Fusion (SAPF) modules combined with ResNet34.**	**Skin Lesion: The Jaccard index improved by 1.74% over baseline. Competitive on 4 medical datasets.**	**Single-stage context extraction in U-shape structures is insufficient for imbalanced classes.**
Quyen & Kim	**Feature Pyramid Network (FPN) with multi-scale prediction fusion for real-time segmentation.**	**ResNet50 backbone with FPN and a multi-scale prediction fusion module using addition/concatenation.**	**Cityscapes: 77.9% mIoU at 62 FPS, outperforming standard FPN (71.3%).**	**Focuses on real-time trade-offs; standard FPNs often lack high-resolution detail retention.**

Literature Review

Author	Description	Methodology	Results	Limitations
Liu et al.	**Multi-scale pyramid residual weight network (LYWNet) for medical image fusion.**	**Multi-scale pyramid residual weight blocks with a feature distillation fusion strategy.**	**SPECT-MRI: SSIM 0.5592. CT-MRI: SSIM 0.5376. High PSNR across modalities.**	**Excessive scale sums in pyramids can lead to high computational cost (addressed by reducing layers).**
Zhang et al. International Journal of CARS	**Multi-Scale Feature Pyramid Fusion Network (MS-Net) for medical segmentation.**	**Multi-Scale Attention Module (MSAM), Stacked Feature Pyramid Module (SFPM), Feature Perception Module.**	**CHAOS (Liver): 94.54% Dice. ISIC 2018: 86.06% dice. Outperforms U-Net++.**	**SFPM module sacrifices parameters for performance, consuming significant time and memory.**
Xu et al.	**MPCT: Medical image fusion based on multi-scale pyramid convolution and Transformer.**	**Syncytial CNN-Transformer (SCT) module within a multi-scale pyramid structure.**	**MRI-SPECT: PSNR 66.698. Has the fewest parameters among the compared methods.**	**CNNs miss global context, while Transformers miss local details.**
Wang et al. PLoS ONE	**Fine-grained classification network using multi-scale pyramid convolutions.**	**Replaces standard convolutions with PyConv (multi-scale kernels: 9x9 to 3x3) and adds a multi-attention module.**	**CUB-200-2011: 85.92% Top-1 Acc. Stanford Cars: 93.64%. Beats WS-DAN.**	**The proposed method still retains a relatively large number of parameters.**

Research Gaps Identified

- **Loss of High-Frequency Details:** Standard downsampling permanently discards fine boundaries and textures; upsampling cannot fully recover them due to aliasing and Nyquist limits.
- **Single-Scale Frequency Analysis:** Existing FFT-based approaches extract spectral features at only one network depth, missing the multi-scale nature of pathological characteristics.
- **Lack of Explainability:** Frequency-domain models produce decisions hard to interpret — spectral components lack direct spatial correspondence, limiting clinical trust.
- **Feature Fusion Challenges:** Naively combining spatial and frequency features introduces numerical instability due to differing distributions.
- **Cross-Domain Generalisability:** Most methods are evaluated on a single dataset, leaving open the question of cross-modality applicability (MRI vs. radiography).

Problem Statement & Objectives

Addressing the loss of fine-grained high-frequency details due to repeated downsampling by proposing a multi-scale frequency pyramid fusion framework that integrates FFT-based representations from intermediate layers using learnable gating for enhanced discrimination and interpretability.

Objectives:

1. **Phase 1:** Dual-frequency feature fusion with 9-channel spectral representation and Score-CAM explainability.
2. **Phase 2:** Multi-scale frequency pyramid with Frequency Attention Gates at intermediate scales, FPN with softmax fusion, and hierarchical frequency-aware Score-CAM explainability.

Datasets Used

Dataset 1: Brain Tumor MRI (4-class)

Class	Train	Test	Total
Glioma	1,321	400	1,721
Meningioma	1,339	400	1,739
No Tumor	1,595	400	1,995
Pituitary	1,457	400	1,857
Total	**5,712**	**1,600**	**7,312**

Dataset 2: Chest X-Ray Pneumonia (Binary)

Class	Train	Test	Total
Normal	1,341	234	1,575
Pneumonia	3,875	390	4,265
Total	**5,216**	**624**	**5,840**

Why two datasets? Tests cross-modality generalisability: MRI vs. radiography, multi-class vs. binary, focal lesions vs. diffuse opacities.

Spatial-to-Frequency Domain Conversion

2D Discrete Fourier Transform per channel ($c \in \{R, G, B\}$):

$$F_c(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f_c(x,y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Decomposition into 3 components per channel:

- **Log-Magnitude:** $L_c(u,v) = \ln(\max(|F_c|, \epsilon) + \epsilon)$ (spectral energy)
- **Cosine Phase:** $P_c^{\cos}(u,v) = \cos(\Phi_c(u,v))$ (spatial structure)
- **Sine Phase:** $P_c^{\sin}(u,v) = \sin(\Phi_c(u,v))$ (spatial localization)

9-Channel Frequency Tensor:

$$\mathrm{X}_{\text{freq}} = \left[L_R, P_R^{\cos}, P_R^{\sin}, L_G, P_G^{\cos}, P_G^{\sin}, L_B, P_B^{\cos}, P_B^{\sin}\right] \in \mathbb{R}^{9 \times H \times W}$$

Backbone: Modified CNN first conv layer: $\mathbb{R}^{9 \times H \times W} \rightarrow \mathbb{R}^{64 \times \frac{H}{2} \times \frac{W}{2}}$
Weight Init: Pretrained weights for channels 0–2; scaled Kaiming ($\times 0.1$) for channels 3–8.

Dual-Frequency Feature Fusion (DFFF)

Operates on Layer 4 activation maps $\mathrm{A} \in \mathbb{R}^{B\times C\times h\times w}$:

Step 1: Second-level FFT on activations: $M_k^{\mathrm{feat}} = |\mathrm{FFT}_{2\mathrm{D}}(A_k)|$

Step 2: Separate Batch Normalization:

$$\tilde{\mathrm{A}}_{\mathrm{spatial}} = \mathrm{BN}_{\mathrm{spatial}}(\mathrm{A}), \qquad \tilde{\mathrm{A}}_{\mathrm{freq}} = \mathrm{BN}_{\mathrm{freq}}(\mathrm{A}_{\mathrm{freq}})$$

Step 3: Concatenate + Channel Attention (SE-style):

$$\mathrm{C} = [\tilde{\mathrm{A}}_{\mathrm{spatial}};\ \tilde{\mathrm{A}}_{\mathrm{freq}}], \quad \mathrm{s} = \sigma(\mathrm{W}_2 \cdot \delta(\mathrm{W}_1 \cdot \mathrm{GAP}(\mathrm{C})))$$

Step 4: Learnable Domain Fusion:

$$\mathrm{A}_{\mathrm{fused}} = \alpha \cdot \hat{\mathrm{A}}_s + \beta \cdot \hat{\mathrm{A}}_f$$

where $\alpha = \sigma(\alpha_{\mathrm{raw}})$, $\beta = \sigma(\beta_{\mathrm{raw}})$ (init: $\alpha \approx 0.7$, $\beta \approx 0.3$)

Design Rationale: Separate BN prevents distribution mismatch; sigmoid-gated weights allow gradual frequency contribution; SE attention selects salient channels across both domains.

Hierarchical Frequency-Spatial Attention (HFSA)

Asymmetric Design: Frequency processing at 3 scales:

Frequency Attention Gate (FAG): at Layer 2 (28×28) and Layer 3 (14×14):

1. Compute FFT magnitude: $x_{freq} = |FFT_{2D}(x)|$
2. Lightweight convolutional gate:

$$g = \sigma(\mathrm{Conv}_{1\times1}(\delta(\mathrm{BN}(\mathrm{Conv}_{3\times3}(x_{freq})))))$$

3. Residual-style gating:

$$x_{out} = x \odot (1 + \alpha \cdot g), \qquad \alpha_{init} = 0.1$$

Full DFFF: at Layer 4 (7×7):

- Complete spatial-frequency fusion (as in Phase 1) at the most abstract level.

Why asymmetric? At high resolutions (L2, L3), lightweight gating avoids instability. At L4 (7×7), full dual-domain fusion is computationally negligible and appropriate for abstract features.

Feature Pyramid Network & Learnable Fusion

Top-Down Pathway:

$$\mathrm{p}_4 = \mathrm{DFFF}(\mathrm{x}_4) \in \mathbb{R}^{B\times 2048\times 7\times 7}$$
$$\mathrm{p}_3 = \mathrm{Upsample}(\mathrm{p}_4, 14 \times 14) + \mathrm{Lat}_3(\mathrm{FAG}_3(\mathrm{x}_3))$$
$$\mathrm{p}_2 = \mathrm{Upsample}(\mathrm{p}_3, 28 \times 28) + \mathrm{Lat}_2(\mathrm{FAG}_2(\mathrm{x}_2))$$

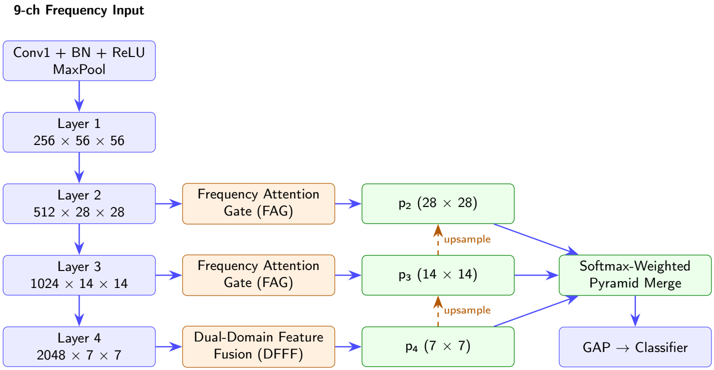
Adaptive Pyramid Fusion (Softmax-Weighted):

$$\mathrm{w} = \mathrm{softmax}(\boldsymbol{\ell}), \quad \boldsymbol{\ell} \in \mathbb{R}^3 \text{ (learnable logits)}$$

$$\mathrm{M}_{\mathrm{merged}} = w_1 \cdot \mathrm{Pool}(\mathrm{p}_2) + w_2 \cdot \mathrm{Pool}(\mathrm{p}_3) + w_3 \cdot \mathrm{p}_4$$
$$\mathrm{M}_{\mathrm{out}} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times 3}(\mathrm{M}_{\mathrm{merged}})))$$

Key Property: Softmax ensures $\sum w_i = 1$. The model adaptively learns scale contributions, coarse features for large lesions, and fine-grained features for diffuse/subtle pathologies.

Proposed MSPF Architecture



Multi-Scale Score-CAM Explainability

Score-CAM Pipeline (Gradient-Free):

1. Extract fused activations from HFPN: $\mathrm{A}_{\mathrm{fused}} \in \mathbb{R}^{1\times 2048\times 7\times 7}$
2. Upsample each channel k to 224×224 and normalize to $[0, 1]$
3. Apply as soft mask: $X_k^* = X_{\mathrm{freq}} \odot \bar{A}_k$
4. Forward pass $\rightarrow$ target-class confidence: $s_k = \mathrm{softmax}(f(X_k^*))_t$
5. Weighted CAM:

$$\mathrm{CAM}(u, v) = \mathrm{ReLU}\left(\sum_{k=1}^{C} s_k \cdot A_k(u, v)\right)$$

Frequency-to-Spatial Mapping:

$$I_{\mathrm{sal}} = \mathrm{IFFT}_{2\mathrm{D}}\left(\mathrm{CAM} \cdot M \cdot e^{j\Phi}\right)$$

Post-Processing: Gaussian smoothing ($\sigma = 2.5$) $\rightarrow$ 40th percentile thresholding $\rightarrow$ morphological close/open $\rightarrow$ power-law enhancement ($\gamma = 0.7$) $\rightarrow$ spatial overlay.

Training Setup

Optimiser & LR:

- AdamW, weight decay 5×10^{-4}
- Differential LR: backbone $\times 0.01$, new modules $\times 0.5$
- Cosine annealing with warm restarts ($T_0 = 15$, $T_{\mathrm{mult}} = 2$)

Stability:

- Mixed Precision (AMP)
- Gradient clipping (max norm 1.0)
- Label smoothing $\epsilon = 0.1$
- Early stopping (patience=15)

Augmentation:

- Random horizontal/vertical flip
- Random rotation: $\pm 20^\circ$
- Color jitter: (brightness, contrast, saturation)
- Random affine (translation and scale)
- ImageNet normalization
- FFT conversion (9-channel)

Hardware: NVIDIA GPU (16 GB VRAM), CUDA-enabled

Software: PyTorch + torchvision, CuDNN Benchmark Mode

Results

Multi-Backbone Results (Test Acc %):

Backbone	Brain MRI	Chest X-Ray
ResNet-50	82.75	81.41
MobileNetV2	79.44	79.17
DenseNet-121	**85.63**	**84.45**
EfficientNet-B0	83.88	82.37
VGG-19	78.19	77.56

Key Findings:

- DenseNet-121 achieves best performance on both datasets (**85.63%** / **84.45%**)
- Brain Tumor: No Tumor class 98.75% accuracy; Pituitary 93.75%; Glioma 62.25% (hardest due to diffuse morphology)
- Chest X-Ray: Pneumonia recall **94.62%** (high sensitivity); Normal recall 59.40% (class imbalance effect)
- Framework generalizes across 5 diverse CNN architectures and 2 imaging modalities

Loss and Accuracy Graphs

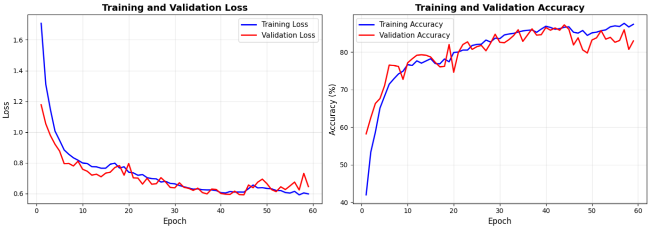


Figure 1: Brain Tumor MRI dataset.

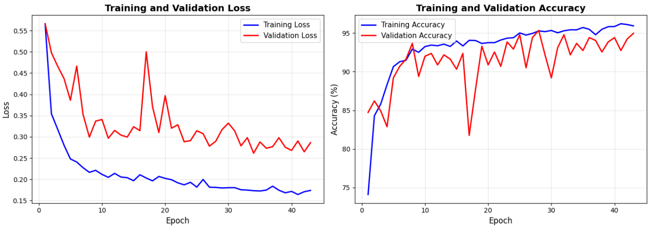


Figure 2: Chest X-Ray Pneumonia dataset.

CAM Visualisations

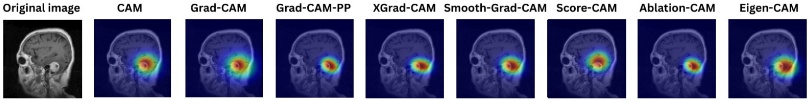


Figure 3: CAM variants visualisation for Brain Tumor MRI

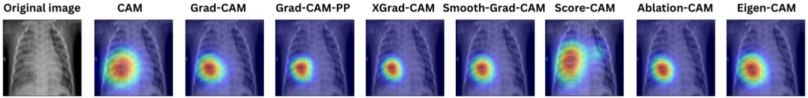


Figure 4: CAM variants visualisation for Chest X-ray Pneumonia

Conclusion

Summary:

- Proposed a two-phase framework: **DFFF** (Phase 1) $\rightarrow$ **MSPF with HFSA** (Phase 2) for frequency-domain medical image classification with explainability.
- 9-channel spectral encoding, asymmetric multi-scale frequency processing (FAG at L2/L3 + full DFFF at L4), and FPN with learnable softmax fusion.
- Validated across **5 CNN backbones** and **2 medical imaging modalities**: Brain Tumor MRI (85.63%) and Chest X-Ray Pneumonia (84.45%).
- Frequency-domain Score-CAM produces clinically interpretable saliency maps.

Key Contributions:

- 9-channel FFT spectral encoding (magnitude + sin/cos phase per RGB channel)
- DFFF module with channel attention & learnable domain weights
- HFSA architecture with FAG + FPN + learnable softmax fusion
- Frequency-domain Score-CAM with spatial mapping
- Validation on two medical imaging benchmarks across 5 CNN backbones

Future Works

- **Multi-Resolution Frequency Pyramid:** Execute frequency analysis across multiple resolutions and stitch via attention-based fusion.
- **Spatial-Frequency Co-Attention:** Implement bidirectional cross-attention between spatial and frequency streams.
- **Extended Clinical Validation:** Test on larger, multi-institutional datasets (e.g., retinal, histopathology) for real-world robustness.
- **Multi-Scale CAM Framework:** Develop level-specific interpretations for granular, localized clinical explanations.
- **Domain-Adaptive Frequency Encoding:** Explore modality-specific spectral transforms (e.g., wavelets) tailored to distinct imaging characteristics.

Conference Submission

Conference Paper Submitted

Magnitude and Phase Aware Spectral Convolutional Neural Networks: Explainable Frequency-Domain Classification

Submitted to International Conference on Computer Vision and Image Processing (CVIP) 2026, NIT Calicut (Springer).

Paper covers:

- A spectral learning framework utilizing a 9-channel FFT representation (magnitude, sine, and cosine phase) with gradient-stabilised feature fusion.
- A frequency-aware Class Activation Mapping (CAM) approach providing precise spatial interpretability for frequency-domain predictions.
- Extensive evaluations across medical and non-medical image datasets using multiple CNN architectures to validate robustness.

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